



CS & IT ENGINEERING



Algorithms

Divide & Conquer

Lecture No.- 07



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Quick Sort Questions
Matrix Multiplication

Topics to be Covered



Topic

Topic

Topic

TC analysis using

Masters Method



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.



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TTTe



Topic : Divide & Conquer

Solving DnC Recurrence

1. Back Substitution method →
 - (i) Value of recurrence
 - (ii) TC of recurrence → $O()$

Min-max prob :

$$\text{Value : } \left\lceil \frac{3n}{2} - 2 \right\rceil$$

$$\text{TC} \rightarrow O(n)$$

Universal method

2. Master's method/Theorem →
 - (i) Only gives TC but not value
 - (ii) Works only in specific cases

Shortcut

(3 cases)

3. Recurrence tree approach :
 - (i) Asymmetric DnC recurrence



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Master's method : Only applicable for

$$T(n) = a * T(n/b) + F(n)$$

Such that $a \geq 1$, $b > 1$, $F(n) = +ve$

→

Symmetric DnC

Case 1 : If $F(n)$ is $O(n^{\log_b a - E})$, for same $E > 0$,

Then

$$T(n) = \theta(n^{\log_b a})$$

~~$\leq n^x - 100$~~

$\log_b a$



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Case 2 : If $F(n)$ is $\theta(n^{\log_b a} * (\log n)^{\underline{K}})$ for some K , such that

(a) $K \geq 0$, then $T(n) = \theta(n^{\log_b a} * (\log n)^{\underline{K+1}})$

(b) $K = -1$, then $T(n) = \theta(n^{\log_b a} * \underline{\log(\log n)})$



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Case 3 : If $f(n)$ is $\underline{\Omega}(n^{\log_b a + E})$

For some $E > 0$

AND

$$a * f(n/b) \leq \delta * f(n),$$

For some $\delta < 1$

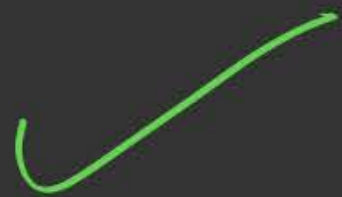
Then,

$$T(n) = \theta(f(n))$$

$$T(n) = 2T(n/2) + n.$$

$$T(n) = aT(n/b) + \bar{f}(n)$$

$$\begin{array}{l} a=2 \\ b=2 \\ \bar{f}(n)=n \end{array}$$



$$\log_b a = \log_2 2 = \textcircled{1}$$

Case 1:-

$$n = O(n^{1-\epsilon}), \text{ some } \epsilon > 0 ?$$

No

Case 2:- $n = O(n' * (\log n)^k)$, some k

a) $k > 0$, $(k=0)$ ✓ Yes

$$T_C = O(n' * (\log n)^{0+1}) \Rightarrow \boxed{O(n \log_2 n)}$$



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#Q. $T(n) = 4T(n/2) + n$

Check master's method form :

$$T(n) = a * T(n/b) + F(n)$$

$$a = 4, b = 2, F(n) = n \Rightarrow$$

$$\log_b a = \log_2 4 = \log_2 2^2 = \underline{2}$$

Valid



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Check Case 1 : If $F(n) = O(n^{2-E})$, some $E > 0$

Is $n = O(n^{2-E})$, $E > 0$

$$E = \underline{0.5}, \underline{0.1}, \underline{0.2}, \dots$$

$$n = O(n^{1.5})$$

True ✓

Case 1 satisfied

Hence

$$T(n) = \theta(n^2)$$



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Check Using Back-substitution :

$$T(n) = 4T(n/2) + n$$

$$T(n/2) = 4T(n/2^2) + n/2$$

$$T(n) = 4(4T(n/2^2) + n/2) + n$$

$$= 4^2 T(n/2^2) + 2n + n$$

$$= 4^2 T(n/2^2) + 3n$$

$$= 4^3 T(n/2^3) + 7n$$

.

.

.

$$T(n) = 4^K T(n/2^K) + (2^K - 1)n$$

$$n/2^K = 1$$

$$2^K = n \quad \Rightarrow \quad 4^K = n^2$$

$$T(n) = n^2 T(1) + (n - 1)n$$

$$= n^2 * c + n^2 - n$$

$$T(n) = \theta(n^2)$$



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#Q. $T(n) = 2T(n/2) + n \log n$

Check : $a = 2, b = 2, F(n) = n \log n$

$$\log_b a = \log_2 2 = 1$$

Check case 1 :

If $F(n) = O(n^{\log_b a - E})$,

$$n \log n = O(n^{1-E}),$$

$\Rightarrow$ Valid

$$E > 0$$

some $E > 0$?

Fails

$$(n \log n \leq n^{1-E})$$



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Check Case 2 :

$$\text{If } F(n) = \theta(n^{\log_b a} * (\log n)^K)$$

Case 2 : a) $K \geq 0$

$$n \log n = \theta(n^1 * (\log n)^K) \rightarrow \text{True for } \underline{K = 1}$$

Case (a) satisfied.

$$\begin{aligned} \text{hence, } T(n) &= \theta(n^1 * (\log n)^{K+1}), & K = 1 \\ &= \theta(n * (\log n)^{1+1}) \end{aligned}$$

$$T(n) = \theta(n * (\log n)^2)$$



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#Q. $T(n) = T(n/3) + n$

A

$$T(n) = \theta(n \log n)$$

B

$$T(n) = \theta(n)$$

C

$$T(n) = \theta(n^2)$$

D

$$T(n) = \theta(\log n)$$

70.8%



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Validity :

$$a = 1, b = 3, F(n) = n$$

$$\log_b a = \log_3 1 = 0$$



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Case 1 :

If $F(n) = O(n^{\log_b a - E}), \quad E > 0$
 $n = O(n^{0 - E}), \quad \text{some } E > 0$

Handwritten green annotations:
A green line is drawn under the expression $n = O(n^{0 - E})$.
A green arrow points from the underlined expression down to the word "deus".

Fails

Case 2 :

$$n = \theta(n^0 * (\log n)^K), \quad \text{some } K$$

(a) $K \geq 0 \quad \rightarrow \quad \text{Fails}$

(b) $K = -1 \quad \rightarrow \quad \text{Fails}$



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Case 3 :

If $F(n) = \Omega(n^{\log_b a + E})$, ^{Some} $E > 0$

$$n = \Omega(n^{0+E}), \quad E > 0?$$

$$n = \Omega(n^{0.1}) \quad \checkmark$$

$$E = 0.1, 0.5, 0.2, \dots n \geq n^E, E > 0$$

and

$$a * F(n/b) \leq \delta * F(n),$$

some $\delta < 1$

$$1 * n/3 \leq \cancel{1/3 * n}, \quad \delta * n$$

some $\delta < 1$

let take $\delta = 1/3$

$$1 * n/3 \leq 1/3 * n \quad \checkmark$$

$\rightarrow$

True

$\rightarrow$ Here case 3 holds true.

Hence $T(n) = \theta(F(n))$

$\Rightarrow$

$T(n) = \theta(n)$

HW $\rightarrow$ Solve by Back Sub



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#Q. $T(n) = 9T(n/3) + n^{2.5}$

A

$$T(n) = \theta(n)$$

B

$$T(n) = \theta(n\sqrt{n})$$

C

$$T(n) = \theta(n \log n)$$

D

$$T(n) = \theta(n^2\sqrt{n})$$

68.75%

$$\begin{aligned} n^{2.5} &= n^2 \times n^{0.5} \\ &= n^2 \times \sqrt{n} \end{aligned}$$



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$$a = 9, b = 3, F(n) = n^{2.5}$$

→ Valid

$$\log_b a = \log_3 9 = 2$$

$$n^2 * \sqrt{n}$$

$$n^2 * n^{0.5}$$

$$n^{(2 + 0.5)}$$

$$n^{2.5}$$



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Case 1 : Is $n^{2.5} = O(n^{2-E})$,

$$n^{2.5} \leq n^{2-E} ?$$

Some $E > 0$

→ Not possible

→ Fails

Case 2 : Is $n^{2.5} = \theta(n^2 * (\log n)^K)$,

For some K

(a) $K \geq 0 \rightarrow$ Fail

(b) $K = -1 \rightarrow$ Fail

$$n^{2.5} = n^2 (\log n)^K$$

$$\sqrt{n} = (\log n)^K$$

→ Fails

$$n^{2.5} = \theta(n^2 (\log n)^K)$$

$$\sqrt{n} = \theta((\log n)^K) ?$$



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Case 3: Is $n^{2.5} = \Omega(n^{2+E})$

$$n^{2.5} > n^{2+E},$$

→ yes

some $E > 0$

$$E = 0.1, 0.2, 0.33, \dots$$

and

$$a * F\left(\frac{n}{b}\right) \leq \delta * F(n)$$

For some $\delta < 1$

$$F(n) = n^{2.5}$$

$$a * \left(\frac{n}{3}\right)^{2.5} \leq \delta * n^{2.5}$$

$$F(n/3) = (n/3)^{2.5}$$

$$\frac{9}{3^{2.5}} * n^{2.5} \leq \delta * n^{2.5}$$

Case 3 is satisfied.

⇒

$$\delta = 1/\sqrt{3}$$

$$T(n) = \theta(F(n)) = \theta(n^{2.5})$$



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Doubt :

$$~~F(n) = n^{2.5}~~$$

$$F(n) = n^{2.5}$$

$$F\left(\frac{n}{3}\right) \rightarrow \left(\frac{n}{3}\right)^{2.5}$$

$$n \rightarrow \frac{n}{3}$$

Validate all the D_nC recurrences that we have learned so far using Master's method.



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1. Min-Max Problem :

$$T(n) = 2T(n/2) + 2$$

→

$$\text{Value of Rec} \Rightarrow \left(\frac{3n}{2} - 2 \right)$$

$$\text{TC} \rightarrow O(n)$$

$$\rightarrow a = 2, b = 2, F(n) = 2 \text{ (+ve)}$$

→

Valid

$$\log_b a$$

$$\log_2 2 = 1$$

Case 1 : Is

$$2 = O(n^{1-E}),$$

$$E > 0$$

Yes

$$E = 0.5, 0.1, 0.2, \dots \text{ True}$$

Hence

$$T(n) = \theta(n^{\log_2 2}) = \theta(n)$$

2) Binary Search

$$T(n) = T(n/2) + C$$





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2. Binary Search :

$$T(n) = \theta(\log n)$$

$$O(\log n)$$

$$T(n) = T(n/2) + C$$

$$\rightarrow a = 1, b = 2, F(n) = C \rightarrow \text{Valid } \log_b a = \log_2 1 = 0$$

Check case 1

$$\text{Is } F(n) = O(n^{\log_b a - E})$$

Some $E > 0$

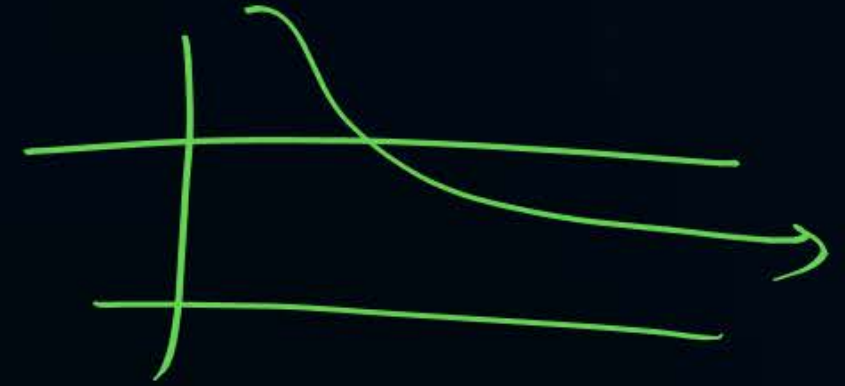
$$C = O(n^{0 - E})$$

Some $E > 0$

$\rightarrow$ Fails

$$n^{-E}, E > 0$$

Decreasing function





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Check case 2 :

Is $F(n) = \theta(n^{\log_b a} * (\log n)^K)$, some K

(a) $K \geq 0$

(b) $K = -1$

$$C = \theta(n^0 * (\log n)^K) ?$$

If $K = 0$, $C = \theta(n^0 * (\log n)^0)$

$$C = \theta(1) \rightarrow \text{True}$$

Hence case 2 (a) holds

$$T(n) = \theta(n^{\log_b a} * (\log n)^{K+1})$$

$$= \theta(n^0 * (\log n)^{0+1})$$

$$T(n) = \theta(\log n)$$

3)

Merge Sort

$$T(n) = 2T(n/2) + n$$



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3. Merge Sort : $TC = \theta(n \log n)$

$$T(n) = 2T(n/2) + n$$

Check $a = 2, b = 2, F(n) = n \rightarrow$ Valid

$$\log_b a = \log_2 2 = 1$$

Check case 1 :

Is $F(n) = O(n^{\log_b a - E})$ some $E > 0$

$n = O(n^{1-E}),$ some $E > 0 \rightarrow$ Fails



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Check case 2 :

Is $F(n) = \theta(n^{\log_b a} * (\log n)^K)$, Some K

(a) $K \geq 0$

(b) $K = -1$

$K = 0,$ $n = \theta(n^1 * (\log n)^0)$
 $n = \theta(n)$

→ True

Case holds for $K = 0$

$$T(n) = \theta(n^{\log_b a} * (\log n)^{K+1})$$
$$= \theta(n^1 * (\log n)^{0+1}) = \theta(n \log n)$$

4) Quick Sort

✓ B.C TC $\Rightarrow T(n) = 2T(n/2) + n$

W.C TC $\Rightarrow T(n) = T(n-1) + n$

5) Matrix multiplication

1) Simple DnC $\Rightarrow T(n) = 8 * T(n/2) + O(n^2)$

2) Strassen DnC $\Rightarrow T(n) = \textcircled{7} * T(n/2) + O(n^2)$



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4. Matrix Multiplication :

(D_nC approach) → $\theta(n^3)$

$$T(n) = 8T(n/2) + n^2$$

$$a = 8, b = 2, F(n) = n^2,$$

$$\log_b a = \log_2 8 = 3$$

Check case 1 :

$$\text{Is } F(n) = O(n^{\log_b a - E}),$$

some $E > 0$

$$\text{Is } n^2 = O(n^{3-E}),$$

some $E > 0$?

→ holds

$E = 0.1, 0.2, \dots$ Yes.

$$\text{Hence, } T(n) = \theta(n^{\log_b a}) = \theta(n^3)$$



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5. Strassen's D_nC Matrix Multiplication :

$$T(n) = 7T(n/2) + n^2$$

$$a = 7, b = 2, F(n) = n^2, \rightarrow$$

$$\text{Valid } \log_b a = \log_2 7 = 2.81$$

Check case 1 :

$$\text{Is } F(n) = O(n^{\log_b a - E}),$$

some $E > 0$

$$\text{Is } n^2 = O(n^{2.81 - E}),$$

some $E > 0$?

$\rightarrow$ True

$E = 0.5, 0.4, \dots$

$$\text{Hence, } T(n) = \theta(n^{\log_b a}) = \theta(n^{2.81})$$

Hw:- 1) $T(n) = 6T(n/3) + n^2 \log n$

2) $T(n) = 4T(n/2) + \log n$



2 mins Summary



Topic

Topic

Topic

Topic

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Masters Method



THANK - YOU